



DIPARTIMENTO DI ELETTRONICA,
INFORMAZIONE E BIOINGEGNERIA

Politecnico di Milano

Machine Learning (Code: 097683)

January 27, 2022

Surname:

Name:

Student ID:

Row:

Column:

Time: 2 hours

Maximum Marks: 33

- The following exam is composed of **8 exercises** (one per page). The first page needs to be filled with your **name, surname and student ID**. The following pages should be used **only in the large squares** present on each page. Any solution provided either outside these spaces or **without a motivation** will not be considered for the final mark.
- During this exam you are **not allowed to use electronic devices**, such as laptops, smartphones, tablets and/or similar. As well, you are not allowed to bring with you any kind of note, book, written scheme, and/or similar. You are also not allowed to communicate with other students during the exam.
- The first reported violation of the above-mentioned rules will be annotated on the exam and will be considered for the final mark decision. The second reported violation of the above-mentioned rules will imply the immediate expulsion of the student from the exam room and the **annulment of the exam**.
- You are allowed to write the exam either with a pen (black or blue) or a pencil. It is your responsibility to provide a readable solution. We will not be held accountable for accidental partial or total cancellation of the exam.
- The exam can be written either in **English** or **Italian**.
- You are allowed to withdraw from the exam at any time without any penalty. You are allowed to leave the room not early than half the time of the duration of the exam. You are not allowed to keep the text of the exam with you while leaving the room.
- **Three of the points will be given on the basis on how quick you are in solving the exam.** If you finish earlier than 45 min before the end of the exam you will get 3 points, if you finish earlier than 30 min you will get 2 points and if you finish earlier than 15 min you will get 1 point (the points cannot be accumulated).
- The box on Page 10 can only be used to complete the Exercises 7, and/or 8.

Ex. 1	Ex. 2	Ex. 3	Ex. 4	Ex. 5	Ex. 6	Ex. 7	Ex. 8	Time	Tot.
/ 7	/ 7	/ 2	/ 2	/ 2	/ 2	/ 4	/ 4	/ 3	/ 33

Exercise 1 (7 marks)

Describe Support Vector Machines (SVMs) for supervised classification problems. In particular explain how do they work, their strengths and weaknesses.

Exercise 2 (7 marks)

Describe the Thomson Sampling algorithm for multi-armed bandit problems.

Exercise 3 (2 marks)

Consider the following snippet of code taking as input an $N \times M$ matrix X of data points and an N -dimensional vector y of binary targets, where N is the number samples, M is the number of features. Are the subsequent statements true or false? Provide motivations for your answers.

```
1  import numpy as np
2  classifier = LogisticRegression()
3  classifier.fit(X, y)
4  yhat = classifier.predict(X)
5  accuracies = [sum(yhat == y) / N]
6  for i in range(M):
7      Xi = np.delete(X, i, axis=1)
8      classifier.fit(Xi, y)
9      yhat = classifier.predict(Xi)
10     accuracies.append(sum(yhat == y) / N)
```

1. This snippet of code implements a portion of a well-known model selection procedure. (*Note*: in any case, specify the procedure and briefly describe how to complement the code).
2. After having run the reported snippet, one should keep the classifier that led to the maximum value in the list of accuracies.
3. The classifier trained at line 3 is likely to suffer a larger bias than the classifiers trained at line 8.
4. The time of computation we need to run the snippet of code scales linearly with M .

1. TRUE, the code implements the first iteration of a typical *backward feature selection* procedure, which is used for model selection. However, some aspects are missing in the snippet, such as the computation of the accuracy on a validation set or cross-validation.
2. FALSE, model selection cannot be performed on the training accuracy, otherwise we would always select the model that keeps all the M features.
3. FALSE, the classifier trained on the full set of features is likely to suffer a larger variance and a lower bias w.r.t. the classifiers trained on a subset of the features.
4. FALSE, whereas we need to run the loop 6-10 exactly M times, the time of computation we need to run the fit function of the classifier grows with M as well, being the logistic regression a parametric method.

Exercise 4 (2 marks)

Tell whether the following statements about sequential decision problems are true or false. Motivate your answers.

1. The optimal value function alone is sufficient to compute the optimal policy.
2. The sparsity of the reward can be a major problem for on-policy RL algorithms.
3. The value function may decrease on some steps of Policy Iteration, even if, in the end, the algorithm outputs the optimal one.
4. As many policies can be optimal, there can be multiple optimal value functions in an MDP.

1. FALSE, the optimal policy is the greedy policy w.r.t. the optimal Q-function. If I have the optimal value function, but not the one-step dynamics, I cannot recover the optimal Q-function. Moreover, since I do not know the model, I cannot use DP techniques, but I have to use RL algorithms.
2. TRUE: Despite reward sparsity is a general issue for RL algorithms, an on-policy algorithm might not be able to explore enough to reach states with higher reward, while an off-policy algorithm might be more aggressive in the exploration.
3. FALSE, the policy improvement theorem guarantees that, at every step, the new policy is better than the previous one.
4. FALSE, the only optimal value function is the unique fixed point of the Bellman Optimal Operator, and all optimal policies share the same value function.

Exercise 5 (2 marks)

A company selling glasses called *TenebrisOptica* ask us to provide a method to estimate the total amount of sales (in terms of number of glasses sold) at the end of the month. The business unit is expecting to have a daily update of this estimate.

1. Do you think that it is appropriate to use an ML approach to solve the previous problem? Which assumptions are required to allow the use of ML for the above application.
2. Assuming that the above assumptions are satisfied, model the above problem using a Machine Learning approach. Suggest both the data one needs and a proper ML method to solve it.
3. Assuming that the above assumption does not hold, what is the problem setting we are in?

1. A crucial assumption required for the application of ML is that the samples are i.i.d. In this specific application the data may be temporally dependent and the process generating data may change over time, therefore the data are not independent nor identically distributed. However, considering a large enough window of data and providing prediction over a time short time period, i.e., so that the nonstationarity is not significant, one might apply ML techniques.
2. The problem is a regression one, where the input are information about past sales, product details, and user feedbacks and the output is the number of sales at the end of the month. The model should take into account the temporal distance from the current day to the end of the month. A possible method to solve the problem is constituted by Linear Regression or Gaussian Process Regression.
3. In the case we cannot rely on the i.i.d. assumption we are in a time series prediction setting. The environment we are considering is either a stochastic nonstationary one or an adversarial one, if we assume that the competitors are significantly affecting the process.

Exercise 6 (2 marks)

Tell if the following statements about learning theory are true or false. Provide adequate motivations for your answers.

1. We can expect all the learning algorithms to perform equally bad on a given learning concept.
2. In the theory of PAC learning, the value of ϵ controls the probability of incurring in a generalization loss greater than δ on the target concept.
3. The VC dimension of an hypothesis space with infinite cardinality cannot be finite.
4. The VC dimension of a linear classifier in a 1-dimensional space is exactly 2.

1. FALSE, according to the No Free Lunch theorem we can expect all the learning algorithms to perform equally bad on the *average* over the concepts. Instead, the specific structure of a given concept can favor an algorithm over another.
2. FALSE, the opposite is true. The value of δ upper bounds the probability of training a model with a generalization loss greater than ϵ .
3. FALSE, the VC dimension is a finer measure of the expressivity of the hypothesis space w.r.t. its cardinality, and it can be finite even if the hypothesis space is not.
4. TRUE. Let us take two points on the axis $y = 0$, say $x_1 = a$ and $x_2 = b$ ($a < b$). For every meaningful combination of labels $(+, +)$, $(-, -)$, $(-, +)$, we can always obtain a perfect classifier with a linear model $x < a$, $x > b$, $a < x < b$, respectively. Instead, if we add a third point $x_3 = c$ ($b < c$), we cannot correctly classify the instance $(+, -, +)$.

Exercise 7 (4 marks)

Given the following results obtained by applying linear regression on a dataset:

OLS Regression Results						
2	Dep. Variable:	y	R-squared (uncentered):	0.452		
4	Model:	OLS	Adj. R-squared (uncentered):	0.445		
	Method:	Least Squares	F-statistic:	61.15		
6	Date:	Fri, 21 Jan 2022	Prob (F-statistic):	4.40e-20		
	Time:	15:33:51	Log-Likelihood:	-167.67		
8	No. Observations:	150	AIC:	339.3		
	Df Residuals:	148	BIC:	345.4		
10	Df Model:	2				
	Covariance Type:	nonrobust				
	coef	std err	t	P-value	[0.025	0.975]
14	x1	1.0715	0.124	8.631	0.000	0.826 1.317
16	x2	-1.3546	0.124	-10.912	0.000	-1.600 -1.109

Answer the following questions:

1. Do you think that each feature is individually significant?
2. Do you think that at least one of the features is significant?
3. How much variance is explained by the model?
4. Do you think that the model is sound?

1. YES, according to the p-value, all of them are significant
2. YES: since the F-statistic has a low p-value.
3. According to the R^2 index, only the 45% of the variance has been explained by the model.
4. From the statistical F-test, it seems that the two features are significant. Moreover we are not using too many predictors for the specific dataset (the residuals DF are 148). One may wonder if modifying the feature set may help obtaining a better fit, since the R^2 value is not particularly high.

Exercise 8 (4 marks)

We are given an Heating, Ventilation, and Air Conditioning (HVAC) in which the states are cold (c), medium (m), warm (w) temperature. We can perform three actions: heat (h), refrigerate (r), and do nothing (d). Assume to have the following partial episodes for the HVAC functioning.

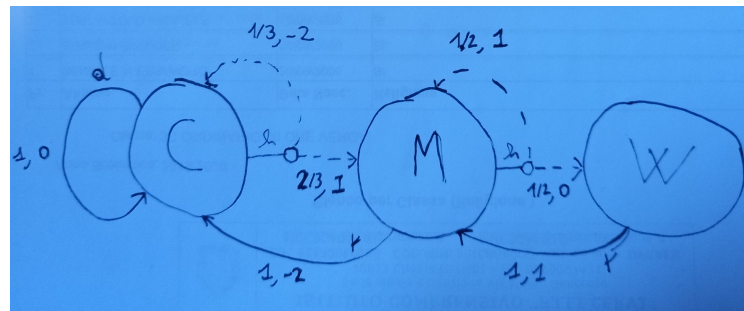
$$(c, d, 0) \rightarrow (c, h, 1) \rightarrow (m, h, 1) \rightarrow (m, h, -1) \rightarrow (w, r, 1) \rightarrow (w, \cdot, \cdot) \rightarrow \dots$$

$$(m, r, -2) \rightarrow (c, h, -2) \rightarrow (c, h, 1) \rightarrow (m, h, 1) \rightarrow (m, h, 1) \rightarrow (w, \cdot, \cdot) \rightarrow \dots$$

where a tuple (S, A, R) correspond to the State, Action, and Reward at a specific time.

1. Model it as an MDP and draw the corresponding graphical representation, specifying the transition probabilities and rewards (estimated from the episodes) for each transition.
2. Can you tell if the reward of this process is stochastic or deterministic? And what about the transitions?
3. Assuming we want to evaluate the performance of the HVAC, tell which kind of problem we are in and suggest a technique to solve it.

1.



2. The transition are stochastic since some of the actions are leading to two different new states, e.g., the action heat (h) in the state cold (c). The reward is stochastic as well, since heating in the medium state provided once the reward 1 and once -1 .
3. This setting suggests it is an MDP prediction problem, either using directly the original episodes (using MC or TD) or one might use the estimated model and use DP techniques to solve it in an exact way, due to the limited dimension of the problem.

